

Relativistic Guidance, Hypersonic Stagnation Thermal Shock, and Matrix Dynamics

Advanced Undergraduate Physics & Aerospace Engineering Seminar

August 10, 2026

Problem Statement

A hyper-velocity exo-/endo-atmospheric interceptor vehicle with variable mass $m(t)$ and inertia tensor $\mathbf{I}_b(t) = \text{diag}(I_{xx}, I_{yy}, I_{zz})$ tracks a evasive target performing high-g maneuvers in 3D space $\mathbb{R}^3$. The vehicle operates within a stratified planetary atmosphere with density $\rho(z) = \rho_0 \exp(-z/H)$ and acoustic speed $a(z) = \sqrt{\gamma R(T_0 - \lambda z)}$.

To intercept ultra-high-speed targets, the flight computer employs a generalized **Augmented Proportional Navigation (APN)** control strategy while managing aero-thermal structural thresholds and variable pitch-yaw-roll dynamic stability.

Part A: Lie Algebra Rotation Kinematics & Vector Cross-Product Operator Mechanics

1. Skew-Symmetric Cross-Product Matrix Operator $\mathbf{S}(\boldsymbol{\omega})$ & Rotation Identities:

Let $\boldsymbol{\omega} = [\omega_x, \omega_y, \omega_z]^T \in \mathbb{R}^3$ be the instantaneous body angular velocity vector. Define the vector cross-product operator matrix $\mathbf{S}(\boldsymbol{\omega}) \in \mathbb{R}^{3 \times 3}$ such that $\mathbf{S}(\boldsymbol{\omega})\mathbf{v} = \boldsymbol{\omega} \times \mathbf{v}$ for any vector $\mathbf{v} \in \mathbb{R}^3$.

- (a) Explicitly construct the 3×3 skew-symmetric matrix $\mathbf{S}(\boldsymbol{\omega})$ and prove the matrix identity:

$$\mathbf{S}(\boldsymbol{\omega})^3 = -\|\boldsymbol{\omega}\|_2^2 \mathbf{S}(\boldsymbol{\omega}) \quad (1)$$

- (b) Using the matrix exponential expansion $\mathbf{R}(\theta, \hat{\mathbf{n}}) = \exp(\theta \mathbf{S}(\hat{\mathbf{n}})) = \sum_{k=0}^{\infty} \frac{\theta^k}{k!} \mathbf{S}(\hat{\mathbf{n}})^k$ for a unit rotation axis $\hat{\mathbf{n}} \in \mathbb{S}^2$, derive **Rodrigues' Rotation Formula** in pure matrix form:

$$\mathbf{R}(\theta, \hat{\mathbf{n}}) = \mathbf{I}_3 + \sin \theta \mathbf{S}(\hat{\mathbf{n}}) + (1 - \cos \theta) \mathbf{S}(\hat{\mathbf{n}})^2 \quad (2)$$

2. Augmented Proportional Navigation (APN) 3D Kinematics: Let $\mathbf{r}(t) = \mathbf{r}_t(t) - \mathbf{r}_v(t)$ be the relative range vector, $R = \|\mathbf{r}\|_2$, $\mathbf{u} = \mathbf{r}/R$ be the Line-of-Sight (LOS) unit vector, and $\boldsymbol{\Omega}_{\text{LOS}} = \mathbf{u} \times \dot{\mathbf{u}}$ be the LOS rotation rate.

- (a) The APN command acceleration $\mathbf{a}_N \in \mathbb{R}^3$ includes target acceleration estimation $\mathbf{a}_t$:

$$\mathbf{a}_N = NV_c(\boldsymbol{\Omega}_{\text{LOS}} \times \mathbf{u}) + \frac{N}{2} \mathbf{a}_{t,\perp} \quad (3)$$

where $V_c = -\dot{R}$ is closing velocity and $\mathbf{a}_{t,\perp} = \mathbf{a}_t - (\mathbf{a}_t \cdot \mathbf{u})\mathbf{u}$ is the component of target acceleration normal to the LOS. Prove that the time derivative of range-rate $\ddot{R}$ under pure engagement geometry ($\mathbf{a}_t = \mathbf{0}$) satisfies:

$$\ddot{R} = R\|\boldsymbol{\Omega}_{\text{LOS}}\|_2^2 - \mathbf{u} \cdot \mathbf{a}_{\text{rel}} \quad (4)$$

Part B: Variable-Mass Aerothermodynamics & Stagnation Enthalpy Integrals

3. Variable-Mass Eulerian Dynamics & Control Deflection Sensitivity: Due to severe solid-rocket motor consumption, the interceptor mass decays linearly as $m(t) = m_0 - \dot{m}_b t$.

- (a) Derive the modified pitch-axis rotational equation of motion accounting for mass loss rate $\dot{m}_b$ and exhaust ejection distance r_e from the center of mass:

$$I_{yy}(t)\dot{q} + \dot{I}_{yy}(t)q = M_{\text{aero}}(\alpha, q, \delta) - \dot{m}_b r_e^2 q \quad (5)$$

- (b) Calculate the instantaneous angular acceleration authority $\frac{\partial \dot{q}}{\partial \delta}$ as an explicit algebraic function of time t , dynamic pressure $q_{\text{dyn}}(t) = \frac{1}{2}\rho(z)V_v^2$, and moment derivative $C_{m\delta}$.

4. Stagnation Point Heat Flux for High-Mach Non-Equilibrium Flow: In extreme Mach regimes ($\mathcal{M} \gg 5$), total stagnation heat transfer rate $q_0(t)$ per unit area obeys the Detra-Kemp-Riddell approximation:

$$q_0(t) = \frac{C}{\sqrt{R_n}} \left(\frac{\rho(z(t))}{\rho_0} \right)^{0.5} \left(\frac{V_v(t)}{V_{\text{ref}}} \right)^{3.15} \quad (6)$$

- (a) Evaluate the exact total thermal dose $Q_{\text{total}} = \int_0^{T_f} q_0(t) dt$ for a vehicle ascending along a linear trajectory $z(t) = z_0 + (V_0 \sin \gamma_0)t$ with quadratic speed deceleration $V_v(t) = V_0(1 - \alpha t)^2$.

Part C: State-Space Controllability Matrix, Rank Conditions, & Lyapunov Stability

- 5. Controllability Matrix Rank & Pole-Placement System Analysis:** Consider the 3rd-order state-space pitch dynamics sub-system $\mathbf{x} = [\Delta\alpha, \Delta q, \Delta\delta]^T \in \mathbb{R}^3$ with finite actuator lag τ_a :

$$\dot{\mathbf{x}} = \begin{bmatrix} -\frac{Z_\alpha}{V_0} & 1 & -\frac{Z_\delta}{V_0} \\ M_\alpha & M_q & M_\delta \\ 0 & 0 & -\frac{1}{\tau_a} \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 0 \\ \frac{1}{\tau_a} \end{bmatrix} u_{\text{cmd}} \quad (7)$$

- (a) Construct the 3×3 Controllability Matrix $\mathcal{C} = [\mathbf{B} \quad \mathbf{AB} \quad \mathbf{A}^2\mathbf{B}]$.
- (b) Prove that the system is completely state controllable ($\text{rank}(\mathcal{C}) = 3$) if and only if $M_\delta + \frac{Z_\delta}{V_0} \left(M_q + \frac{1}{\tau_a} \right) \neq 0$.
- (c) Formulate a symmetric positive-definite matrix $\mathbf{P} = \mathbf{P}^T > 0$ satisfying the **Algebraic Lyapunov Equation**:

$$\mathbf{A}^T \mathbf{P} + \mathbf{P} \mathbf{A} = -\mathbf{Q} \quad (8)$$

for a given $\mathbf{Q} = \mathbf{I}_3$, proving internal asymptotic stability of the unforced system when all eigenvalues $\lambda_i(\mathbf{A})$ satisfy $\text{Re}(\lambda_i) < 0$.

Part D: Mission Narrative, Strategic Mindmap & Mental Framework

Meta-Strategy: Storytelling Mission Overview & Conceptual Deconstruction

Imagine you are the principal flight-dynamics engineer in an high-speed missile defense laboratory. The atmospheric radar arrays trigger an alarm: an ultra-high-speed target is performing evasive maneuvers in the upper atmosphere. Your team must deploy a variable-mass interceptor to track, stabilize, and intercept the target before it escapes.

Your engineering mission unfolds across three main phases:

I. Act I: Lie Algebra & 3D Rotational Kinematics (Part A):

- **What the question is asking:** The flight computer must calculate 3D spatial rotations and track target motion without coordinate singularities or numerical degradation.
- **What we need to do:** Construct the vector cross-product matrix operator $\mathbf{S}(\boldsymbol{\omega})$, prove its cubic identity $\mathbf{S}^3 = -\|\boldsymbol{\omega}\|^2 \mathbf{S}$, expand its matrix exponential to derive ****Rodrigues' Rotation Formula****, and differentiate the relative line-of-sight vector to calculate the relative acceleration rate $\ddot{\mathbf{R}}$.

II. Act II: Variable-Mass Aerodynamics & Thermal Shock (Part B):

- **What the question is asking:** As the solid rocket motor burns fuel rapidly, the vehicle loses mass $m(t)$ and moment of inertia $I_{yy}(t)$. Simultaneously, atmospheric drag heats the nose cone to extreme temperatures.
- **What we need to do:** Modify the Eulerian rotational dynamics equation to include mass-ejection flux rates $\dot{m}_b$, calculate the control fin angular authority $\frac{\partial \dot{q}}{\partial \delta}$, and evaluate the cumulative non-equilibrium stagnation heat flux integral Q_{total} along the flight path.

III. Act III: Control Systems, Controllability, & Stability (Part C):

- **What the question is asking:** Actuator delays (τ_a) and aerodynamic shifts can cause the interceptor to spin out of control. We must verify if our flight computer has full authority to command the missile and prove internal stability.
- **What we need to do:** Form the 3×3 Controllability Matrix $\mathcal{C}$, derive the exact algebraic condition for full rank ($\det(\mathcal{C}) \neq 0$), and solve the algebraic Lyapunov equation $\mathbf{A}^T \mathbf{P} + \mathbf{P} \mathbf{A} = -\mathbf{I}_3$ to prove asymptotic convergence to equilibrium.

1. Mindmap for Part A: Lie Algebra & Matrix Kinematics

Problem 1: Skew-Symmetric Operator $\mathbf{S}(\boldsymbol{\omega})$ & Rodrigues' Formula

- **Mental Picture:** Represent the vector cross product $\boldsymbol{\omega} \times \mathbf{v}$ as a matrix multiplication $\mathbf{S}(\boldsymbol{\omega})\mathbf{v}$. The matrix operator $\mathbf{S}(\hat{\mathbf{n}})$ acts as an infinitesimal generator of 3D rotations around axis $\hat{\mathbf{n}}$.
- **Mathematical Tools:** Matrix algebra, Taylor series expansion, skew-symmetric operator identities.
- **Approach Strategy:**
 1. Construct $\mathbf{S}(\boldsymbol{\omega})$ by matching $\mathbf{S}(\boldsymbol{\omega})\mathbf{v} = \boldsymbol{\omega} \times \mathbf{v}$ component-by-component.
 2. Calculate $\mathbf{S}(\boldsymbol{\omega})^2 = \boldsymbol{\omega}\boldsymbol{\omega}^T - \|\boldsymbol{\omega}\|^2\mathbf{I}_3$. Multiply by $\mathbf{S}(\boldsymbol{\omega})$ once more to establish $\mathbf{S}(\boldsymbol{\omega})^3 = -\|\boldsymbol{\omega}\|^2\mathbf{S}(\boldsymbol{\omega})$.
 3. For Rodrigues' formula, expand $\exp(\theta\mathbf{S}(\hat{\mathbf{n}})) = \sum_{k=0}^{\infty} \frac{\theta^k}{k!}\mathbf{S}(\hat{\mathbf{n}})^k$. Group odd powers ($k = 1, 3, 5, \dots$) to yield $\sin\theta\mathbf{S}$, and group even powers ($k = 2, 4, 6, \dots$) to yield $(1 - \cos\theta)\mathbf{S}^2$.

Problem 2: 3D APN Kinematics

- **Mental Picture:** Interceptor tracking a maneuvering target in 3D space. Range $R(t)$ decreases while line-of-sight unit vector $\mathbf{u}(t)$ rotates at angular rate $\boldsymbol{\Omega}_{\text{LOS}}$.
- **Mathematical Tools:** Vector differential calculus, product rule, orthogonality identities.
- **Approach Strategy:**
 1. Differentiate relative position $\mathbf{r} = R\mathbf{u}$ twice with respect to time to get relative acceleration $\mathbf{a}_{\text{rel}} = \ddot{\mathbf{r}}$.
 2. Take the scalar dot product of the relative acceleration vector equation with $\mathbf{u}$.
 3. Use vector identities $\mathbf{u} \cdot \mathbf{u} = 1 \implies \mathbf{u} \cdot \dot{\mathbf{u}} = 0$ and $\mathbf{u} \cdot \ddot{\mathbf{u}} = -\|\dot{\mathbf{u}}\|^2 = -\|\boldsymbol{\Omega}_{\text{LOS}}\|^2$ to isolate $\ddot{R}$.

2. Mindmap for Part B: Variable-Mass Aerothermodynamics

Problem 3: Variable-Mass Eulerian Dynamics

- **Mental Picture:** Rocket burning solid propellant rapidly. Inertia $I_{yy}(t)$ decreases over time while mass ejection creates reaction momentum damping.
- **Mathematical Tools:** Eulerian angular momentum theorem for open variable-mass systems.
- **Approach Strategy:**
 1. Express pitch angular momentum $H_y = I_{yy}(t)q$ and differentiate using the product rule: $\dot{H}_y = I_{yy}\dot{q} + \dot{I}_{yy}q$.
 2. Include the momentum flux loss term from expelling mass at distance r_e : $\dot{m}_b r_e^2 q$, and equate to external aerodynamic moments M_{aero} .
 3. Isolate angular acceleration $\dot{q}$ and compute partial derivative $\frac{\partial \dot{q}}{\partial \delta} = \frac{q_{\text{dyn}} S_{\text{ref}} c_{\text{ref}} C_{m\delta}}{I_{yy}(t)}$.

Problem 4: Non-Equilibrium Definite Thermal Integral

- **Mental Picture:** Interceptor climbing through air density gradient $\rho(z)$ while velocity decays quadratically. The total thermal dose is the line integral area under the $q_0(t)$ curve.
- **Mathematical Tools:** Variable substitution, exponential integrals, integral scaling.
- **Approach Strategy:**
 1. Express altitude $z(t)$ and velocity $V_v(t)$ as explicit functions of time inside $q_0(t)$.
 2. Factor out all time-invariant physical constants.
 3. Simplify the exponential density factor: $(e^{-z(t)/H})^{0.5} = e^{-z_0/(2H)} e^{-(V_0 \sin \gamma_0 t)/(2H)}$.
 4. Combine exponential air density decay with polynomial velocity decay $(1 - \alpha t)^{6.3}$ and apply variable substitution to form the final definite integral.

3. Mindmap for Part C: Matrix Controllability & Lyapunov Stability

Problem 5: System Controllability & Lyapunov Equations

- **Mental Picture:** Linear state-space model incorporating actuator lag τ_a . We verify if control input u_{cmd} can independently control all three states $(\Delta\alpha, \Delta q, \Delta\delta)$.
- **Mathematical Tools:** Matrix multiplication, determinant rank test, linear matrix equations.
- **Approach Strategy:**
 1. Compute vectors $\mathbf{B}$, $\mathbf{AB}$, and $\mathbf{A}^2\mathbf{B}$ through matrix-vector multiplication.
 2. Construct 3×3 matrix $\mathcal{C} = [\mathbf{B} \quad \mathbf{AB} \quad \mathbf{A}^2\mathbf{B}]$ and compute determinant $\det(\mathcal{C})$.
 3. Set $\det(\mathcal{C}) \neq 0$ to find the exact algebraic condition for full rank controllability.
 4. Set up $\mathbf{A}^T\mathbf{P} + \mathbf{PA} = -\mathbf{I}_3$ for symmetric matrix $\mathbf{P}$ and solve the system of 6 linear algebraic equations for elements p_{ij} to guarantee asymptotic stability.

Part E: Complete Step-by-Step Mathematical Solutions & Detailed Explanations

Problem 1: Skew-Symmetric Operator & Rodrigues' Formula Proof

(a) **Matrix Construction & Operator Cube Identity** $\mathbf{S}(\boldsymbol{\omega})^3 = -\|\boldsymbol{\omega}\|_2^2 \mathbf{S}(\boldsymbol{\omega})$

Let $\boldsymbol{\omega} = [\omega_x, \omega_y, \omega_z]^T$ and $\mathbf{v} = [v_x, v_y, v_z]^T$. The cross product $\boldsymbol{\omega} \times \mathbf{v}$ is:

$$\boldsymbol{\omega} \times \mathbf{v} = \begin{bmatrix} \omega_y v_z - \omega_z v_y \\ \omega_z v_x - \omega_x v_z \\ \omega_x v_y - \omega_y v_x \end{bmatrix} \quad (9)$$

Matching this to matrix-vector product $\mathbf{S}(\boldsymbol{\omega})\mathbf{v}$ yields the explicit skew-symmetric matrix:

$$\mathbf{S}(\boldsymbol{\omega}) = \begin{bmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{bmatrix} \quad (10)$$

Now compute the square $\mathbf{S}(\boldsymbol{\omega})^2 = \mathbf{S}(\boldsymbol{\omega})\mathbf{S}(\boldsymbol{\omega})$ by explicit matrix multiplication:

$$\mathbf{S}(\boldsymbol{\omega})^2 = \begin{bmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{bmatrix} \begin{bmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{bmatrix} \quad (11)$$

$$= \begin{bmatrix} -\omega_y^2 - \omega_z^2 & \omega_x \omega_y & \omega_x \omega_z \\ \omega_x \omega_y & -\omega_x^2 - \omega_z^2 & \omega_y \omega_z \\ \omega_x \omega_z & \omega_y \omega_z & -\omega_x^2 - \omega_y^2 \end{bmatrix} \quad (12)$$

Adding and subtracting $\omega_x^2, \omega_y^2, \omega_z^2$ along the diagonal:

$$\mathbf{S}(\boldsymbol{\omega})^2 = \begin{bmatrix} \omega_x^2 & \omega_x \omega_y & \omega_x \omega_z \\ \omega_x \omega_y & \omega_y^2 & \omega_y \omega_z \\ \omega_x \omega_z & \omega_y \omega_z & \omega_z^2 \end{bmatrix} - (\omega_x^2 + \omega_y^2 + \omega_z^2) \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \boldsymbol{\omega} \boldsymbol{\omega}^T - \|\boldsymbol{\omega}\|_2^2 \mathbf{I}_3 \quad (13)$$

Now multiply $\mathbf{S}(\boldsymbol{\omega})^2$ by $\mathbf{S}(\boldsymbol{\omega})$ to get the matrix cube $\mathbf{S}(\boldsymbol{\omega})^3$:

$$\mathbf{S}(\boldsymbol{\omega})^3 = \mathbf{S}(\boldsymbol{\omega})^2 \mathbf{S}(\boldsymbol{\omega}) = (\boldsymbol{\omega} \boldsymbol{\omega}^T - \|\boldsymbol{\omega}\|_2^2 \mathbf{I}_3) \mathbf{S}(\boldsymbol{\omega}) = \boldsymbol{\omega} \boldsymbol{\omega}^T \mathbf{S}(\boldsymbol{\omega}) - \|\boldsymbol{\omega}\|_2^2 \mathbf{S}(\boldsymbol{\omega}) \quad (14)$$

Evaluating the vector-matrix product $\boldsymbol{\omega}^T \mathbf{S}(\boldsymbol{\omega})$:

$$\boldsymbol{\omega}^T \mathbf{S}(\boldsymbol{\omega}) = [\omega_x \quad \omega_y \quad \omega_z] \begin{bmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{bmatrix} = [0 \quad 0 \quad 0] = \mathbf{0}^T \quad (15)$$

(physically representing $\boldsymbol{\omega} \cdot (\boldsymbol{\omega} \times \mathbf{v}) = 0$).

Therefore, $\boldsymbol{\omega} \boldsymbol{\omega}^T \mathbf{S}(\boldsymbol{\omega}) = \mathbf{0}_{3 \times 3}$, proving:

$$\mathbf{S}(\boldsymbol{\omega})^3 = -\|\boldsymbol{\omega}\|_2^2 \mathbf{S}(\boldsymbol{\omega}) \quad \blacksquare \quad (16)$$

(b) Derivation of Rodrigues' Rotation Formula

For a unit vector $\hat{\mathbf{n}} \in \mathbb{S}^2$, $\|\hat{\mathbf{n}}\|_2 = 1$, so the cube identity simplifies to:

$$\mathbf{S}(\hat{\mathbf{n}})^3 = -\mathbf{S}(\hat{\mathbf{n}}) \quad (17)$$

By induction, higher powers recurse as:

$$\mathbf{S}(\hat{\mathbf{n}})^4 = \mathbf{S}(\hat{\mathbf{n}})^3 \mathbf{S}(\hat{\mathbf{n}}) = -\mathbf{S}(\hat{\mathbf{n}})^2 \quad (18)$$

$$\mathbf{S}(\hat{\mathbf{n}})^5 = -\mathbf{S}(\hat{\mathbf{n}})^3 = \mathbf{S}(\hat{\mathbf{n}}) \quad (19)$$

$$\mathbf{S}(\hat{\mathbf{n}})^6 = -\mathbf{S}(\hat{\mathbf{n}})^2 \quad (20)$$

Now expand the matrix exponential definition $\mathbf{R}(\theta, \hat{\mathbf{n}}) = \exp(\theta \mathbf{S}(\hat{\mathbf{n}}))$:

$$\mathbf{R}(\theta, \hat{\mathbf{n}}) = \mathbf{I}_3 + \theta \mathbf{S}(\hat{\mathbf{n}}) + \frac{\theta^2}{2!} \mathbf{S}(\hat{\mathbf{n}})^2 + \frac{\theta^3}{3!} \mathbf{S}(\hat{\mathbf{n}})^3 + \frac{\theta^4}{4!} \mathbf{S}(\hat{\mathbf{n}})^4 + \frac{\theta^5}{5!} \mathbf{S}(\hat{\mathbf{n}})^5 + \dots \quad (21)$$

Substitute the reduced matrix powers:

$$\mathbf{R}(\theta, \hat{\mathbf{n}}) = \mathbf{I}_3 + \left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \dots \right) \mathbf{S}(\hat{\mathbf{n}}) + \left(\frac{\theta^2}{2!} - \frac{\theta^4}{4!} + \frac{\theta^6}{6!} - \dots \right) \mathbf{S}(\hat{\mathbf{n}})^2 \quad (22)$$

Recognizing the Maclaurin Taylor series expansions for $\sin \theta$ and $\cos \theta$:

$$\sin \theta = \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \dots \quad (23)$$

$$1 - \cos \theta = \frac{\theta^2}{2!} - \frac{\theta^4}{4!} + \frac{\theta^6}{6!} - \dots \quad (24)$$

Substituting these series yields Rodrigues' Rotation Formula in pure matrix form:

$$\mathbf{R}(\theta, \hat{\mathbf{n}}) = \mathbf{I}_3 + \sin \theta \mathbf{S}(\hat{\mathbf{n}}) + (1 - \cos \theta) \mathbf{S}(\hat{\mathbf{n}})^2 \quad \blacksquare \quad (25)$$

Problem 2: 3D APN Kinematics & Range-Rate Acceleration Proof

Relative position is $\mathbf{r} = R\mathbf{u}$. First time derivative gives velocity:

$$\mathbf{v}_{\text{rel}} = \dot{\mathbf{r}} = \dot{R}\mathbf{u} + R\dot{\mathbf{u}} \quad (26)$$

Differentiating again with respect to time gives relative acceleration $\mathbf{a}_{\text{rel}} = \ddot{\mathbf{r}}$:

$$\mathbf{a}_{\text{rel}} = \frac{d}{dt}(\dot{R}\mathbf{u} + R\dot{\mathbf{u}}) = \ddot{R}\mathbf{u} + \dot{R}\dot{\mathbf{u}} + \dot{R}\dot{\mathbf{u}} + R\ddot{\mathbf{u}} = \ddot{R}\mathbf{u} + 2\dot{R}\dot{\mathbf{u}} + R\ddot{\mathbf{u}} \quad (27)$$

Now take the dot product of the entire vector equation with the unit vector $\mathbf{u}$:

$$\mathbf{u} \cdot \mathbf{a}_{\text{rel}} = \mathbf{u} \cdot (\ddot{R}\mathbf{u} + 2\dot{R}\dot{\mathbf{u}} + R\ddot{\mathbf{u}}) = \ddot{R}(\mathbf{u} \cdot \mathbf{u}) + 2\dot{R}(\mathbf{u} \cdot \dot{\mathbf{u}}) + R(\mathbf{u} \cdot \ddot{\mathbf{u}}) \quad (28)$$

We evaluate each scalar dot product term using vector differential identities:

1. Since $\mathbf{u}$ is a unit vector, $\mathbf{u} \cdot \mathbf{u} = 1$.
2. Differentiating $\mathbf{u} \cdot \mathbf{u} = 1$ with respect to time gives $2(\mathbf{u} \cdot \dot{\mathbf{u}}) = 0 \implies \mathbf{u} \cdot \dot{\mathbf{u}} = 0$.
3. Differentiating $\mathbf{u} \cdot \dot{\mathbf{u}} = 0$ with respect to time gives:

$$\dot{\mathbf{u}} \cdot \dot{\mathbf{u}} + \mathbf{u} \cdot \ddot{\mathbf{u}} = 0 \implies \mathbf{u} \cdot \ddot{\mathbf{u}} = -\|\dot{\mathbf{u}}\|_2^2 \quad (29)$$

Since $\mathbf{\Omega}_{\text{LOS}} = \mathbf{u} \times \dot{\mathbf{u}}$ and $\mathbf{u} \perp \dot{\mathbf{u}}$, the norm satisfies $\|\mathbf{\Omega}_{\text{LOS}}\|_2 = \|\mathbf{u}\|_2 \|\dot{\mathbf{u}}\|_2 \sin(90^\circ) = \|\dot{\mathbf{u}}\|_2$. Thus:

$$\mathbf{u} \cdot \ddot{\mathbf{u}} = -\|\mathbf{\Omega}_{\text{LOS}}\|_2^2 \quad (30)$$

Substituting these three scalar results back into the dot product equation:

$$\mathbf{u} \cdot \mathbf{a}_{\text{rel}} = \ddot{R}(1) + 2\dot{R}(0) + R(-\|\mathbf{\Omega}_{\text{LOS}}\|_2^2) = \ddot{R} - R\|\mathbf{\Omega}_{\text{LOS}}\|_2^2 \quad (31)$$

Rearranging to isolate $\ddot{R}$:

$$\ddot{R} = R\|\mathbf{\Omega}_{\text{LOS}}\|_2^2 + \mathbf{u} \cdot \mathbf{a}_{\text{rel}} \quad \blacksquare \quad (32)$$

Problem 3: Variable-Mass Eulerian Dynamics & Deflection Sensitivity

(a) Derivation of Rotational Equation of Motion

From Eulerian mechanics for open systems with variable mass expulsion, the angular momentum about the pitch axis is $H_y = I_{yy}(t)q$. The total rate of change of angular momentum equals external aerodynamic moments minus the momentum flux lost through mass ejection:

$$\frac{dH_y}{dt} + \dot{m}_b r_e^2 q = M_{\text{aero}} \quad (33)$$

Expanding $\frac{d}{dt}(I_{yy}(t)q)$ via the product rule:

$$\frac{d}{dt}(I_{yy}(t)q) = I_{yy}(t)\dot{q} + \dot{I}_{yy}(t)q \quad (34)$$

Substituting yields the variable-mass rotational motion equation:

$$I_{yy}(t)\dot{q} + \dot{I}_{yy}(t)q = M_{\text{aero}}(\alpha, q, \delta) - \dot{m}_b r_e^2 q \quad \blacksquare \quad (35)$$

(b) Instantaneous Deflection Control Sensitivity

Aerodynamic pitching moment is modeled as:

$$M_{\text{aero}} = q_{\text{dyn}}(t)S_{\text{ref}}c_{\text{ref}} \left(C_{m\alpha}\alpha + C_{mq}\frac{qc_{\text{ref}}}{2V_v} + C_{m\delta}\delta \right) \quad (36)$$

Isolating angular acceleration $\dot{q}$ from the motion equation:

$$\dot{q}(t) = \frac{1}{I_{yy}(t)} \left[q_{\text{dyn}}(t)S_{\text{ref}}c_{\text{ref}} \left(C_{m\alpha}\alpha + C_{mq}\frac{qc_{\text{ref}}}{2V_v} + C_{m\delta}\delta \right) - \left(\dot{I}_{yy}(t) + \dot{m}_b r_e^2 \right) q \right] \quad (37)$$

Taking the partial derivative with respect to fin deflection δ :

$$\frac{\partial \dot{q}}{\partial \delta} = \frac{q_{\text{dyn}}(t)S_{\text{ref}}c_{\text{ref}}C_{m\delta}}{I_{yy}(0) - \dot{I}_{yy}t} \quad (38)$$

Problem 4: Stagnation Point Thermal Dose Definite Integral

The heat flux formula is:

$$q_0(t) = \frac{C}{\sqrt{R_n \rho_0}} \sqrt{\rho(z(t))} \left(\frac{V_v(t)}{V_{\text{ref}}} \right)^{3.15} \quad (39)$$

Given:

$$z(t) = z_0 + (V_0 \sin \gamma_0) t \quad (40)$$

$$V_v(t) = V_0(1 - \alpha t)^2 \quad (41)$$

$$\rho(z(t)) = \rho_0 \exp \left(-\frac{z_0 + (V_0 \sin \gamma_0) t}{H} \right) \quad (42)$$

Evaluating $\sqrt{\rho(z(t))}$:

$$\sqrt{\rho(z(t))} = \sqrt{\rho_0} \exp \left(-\frac{z_0}{2H} \right) \exp \left(-\frac{V_0 \sin \gamma_0}{2H} t \right) \quad (43)$$

Evaluating $(V_v(t))^{3.15}$:

$$(V_v(t))^{3.15} = [V_0(1 - \alpha t)^2]^{3.15} = V_0^{3.15} (1 - \alpha t)^{6.3} \quad (44)$$

Combining all constant multipliers into K_0 :

$$K_0 = \frac{C}{\sqrt{R_n}} \exp \left(-\frac{z_0}{2H} \right) \left(\frac{V_0}{V_{\text{ref}}} \right)^{3.15} \quad (45)$$

The definite line integral for total thermal dose Q_{total} is:

$$Q_{\text{total}} = K_0 \int_0^{T_f} \exp \left(-\frac{V_0 \sin \gamma_0}{2H} t \right) (1 - \alpha t)^{6.3} dt \quad (46)$$

Using the integration substitution $u = 1 - \alpha t \implies dt = -\frac{1}{\alpha} du$:

$$Q_{\text{total}} = \frac{K_0}{\alpha} \int_{1-\alpha T_f}^1 \exp \left(-\frac{V_0 \sin \gamma_0}{2H\alpha} (1 - u) \right) u^{6.3} du \quad (47)$$

Factoring out constant exponential terms:

$$Q_{\text{total}} = \frac{K_0}{\alpha} \exp \left(-\frac{V_0 \sin \gamma_0}{2H\alpha} \right) \int_{1-\alpha T_f}^1 \exp \left(\frac{V_0 \sin \gamma_0}{2H\alpha} u \right) u^{6.3} du \quad \blacksquare \quad (48)$$

Problem 5: Controllability Rank & Lyapunov Matrix Derivation

(a) Controllability Matrix Construction $\mathcal{C}$

Given system matrices $\mathbf{A} \in \mathbb{R}^{3 \times 3}$ and $\mathbf{B} \in \mathbb{R}^{3 \times 1}$:

$$\mathbf{A} = \begin{bmatrix} -a & 1 & -b \\ M_\alpha & M_q & M_\delta \\ 0 & 0 & -c \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 \\ 0 \\ c \end{bmatrix} \quad (49)$$

where $a = \frac{Z_\alpha}{V_0}$, $b = \frac{Z_\delta}{V_0}$, and $c = \frac{1}{\tau_a}$.

Compute $\mathbf{AB}$:

$$\mathbf{AB} = \begin{bmatrix} -a & 1 & -b \\ M_\alpha & M_q & M_\delta \\ 0 & 0 & -c \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ c \end{bmatrix} = \begin{bmatrix} -bc \\ M_\delta c \\ -c^2 \end{bmatrix} \quad (50)$$

Compute $\mathbf{A}^2\mathbf{B} = \mathbf{A}(\mathbf{AB})$:

$$\mathbf{A}^2\mathbf{B} = \begin{bmatrix} -a & 1 & -b \\ M_\alpha & M_q & M_\delta \\ 0 & 0 & -c \end{bmatrix} \begin{bmatrix} -bc \\ M_\delta c \\ -c^2 \end{bmatrix} = \begin{bmatrix} abc + M_\delta c + bc^2 \\ -M_\alpha bc + M_q M_\delta c - M_\delta c^2 \\ c^3 \end{bmatrix} \quad (51)$$

Assembling the 3×3 Controllability Matrix $\mathcal{C} = [\mathbf{B} \quad \mathbf{AB} \quad \mathbf{A}^2\mathbf{B}]$:

$$\mathcal{C} = \begin{bmatrix} 0 & -bc & abc + M_\delta c + bc^2 \\ 0 & M_\delta c & -M_\alpha bc + M_q M_\delta c - M_\delta c^2 \\ c & -c^2 & c^3 \end{bmatrix} \quad (52)$$

(b) Rank Condition Proof

The system is fully controllable if and only if $\det(\mathcal{C}) \neq 0$. Expanding $\det(\mathcal{C})$ along the first column:

$$\det(\mathcal{C}) = c \cdot \det \begin{bmatrix} -bc & abc + M_\delta c + bc^2 \\ M_\delta c & -M_\alpha bc + M_q M_\delta c - M_\delta c^2 \end{bmatrix} \quad (53)$$

Factoring c out of each column of the 2×2 minor matrix:

$$\det(\mathcal{C}) = c^3 ((-b)(-M_\alpha b + M_q M_\delta - M_\delta c) - (M_\delta)(ab + M_\delta + bc)) \quad (54)$$

Expanding products inside the bracket:

$$\det(\mathcal{C}) = c^3 (M_\alpha b^2 - bM_q M_\delta + bM_\delta c - abM_\delta - M_\delta^2 - bM_\delta c) \quad (55)$$

Notice the $+bM_\delta c$ and $-bM_\delta c$ terms cancel out cleanly:

$$\det(\mathcal{C}) = -c^3 M_\delta (M_\delta + ab + bM_q) + c^3 M_\alpha b^2 \quad (56)$$

Substituting back $a = \frac{Z_\alpha}{V_0}$, $b = \frac{Z_\delta}{V_0}$, $c = \frac{1}{\tau_a}$, non-singularity ($\det(\mathcal{C}) \neq 0$) establishes full rank 3 if and only if:

$$M_\delta + \frac{Z_\delta}{V_0} \left(M_q + \frac{1}{\tau_a} \right) \neq 0 \quad \blacksquare \quad (57)$$

(c) Lyapunov Equation System Formulation

To prove asymptotic stability via Lyapunov's direct method, we solve $\mathbf{A}^T \mathbf{P} + \mathbf{P} \mathbf{A} = -\mathbf{I}_3$ for symmetric positive-definite matrix $\mathbf{P} = \mathbf{P}^T > 0$:

$$\begin{bmatrix} -a & M_\alpha & 0 \\ 1 & M_q & 0 \\ -b & M_\delta & -c \end{bmatrix} \begin{bmatrix} p_{11} & p_{12} & p_{13} \\ p_{12} & p_{22} & p_{23} \\ p_{13} & p_{23} & p_{33} \end{bmatrix} + \begin{bmatrix} p_{11} & p_{12} & p_{13} \\ p_{12} & p_{22} & p_{23} \\ p_{13} & p_{23} & p_{33} \end{bmatrix} \begin{bmatrix} -a & 1 & -b \\ M_\alpha & M_q & M_\delta \\ 0 & 0 & -c \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \quad (58)$$

Expanding matrix products yields 6 independent linear equations for $p_{11}, p_{12}, p_{13}, p_{22}, p_{23}, p_{33}$:

$$(1, 1) : \quad -2ap_{11} + 2M_\alpha p_{12} = -1 \quad (59)$$

$$(1, 2) : \quad p_{11} + (M_q - a)p_{12} + M_\alpha p_{22} = 0 \quad (60)$$

$$(1, 3) : \quad -bp_{11} + M_\delta p_{12} - (a + c)p_{13} + M_\alpha p_{23} = 0 \quad (61)$$

$$(2, 2) : \quad 2p_{12} + 2M_q p_{22} = -1 \quad (62)$$

$$(2, 3) : \quad -bp_{12} + M_\delta p_{22} + p_{13} + (M_q - c)p_{23} = 0 \quad (63)$$

$$(3, 3) : \quad -2bp_{13} + 2M_\delta p_{23} - 2cp_{33} = -1 \quad (64)$$

Solving this linear system establishes $V(\mathbf{x}) = \mathbf{x}^T \mathbf{P} \mathbf{x} > 0$ as a strict Lyapunov function with time derivative $\dot{V}(\mathbf{x}) = -\mathbf{x}^T \mathbf{I}_3 \mathbf{x} = -\|\mathbf{x}\|_2^2 < 0$, guaranteeing asymptotic convergence $\mathbf{x}(t) \rightarrow \mathbf{0}$. ■